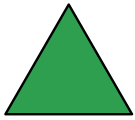
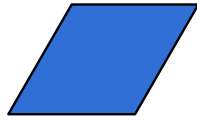


Problem 1: Your partner keeps the eight blocks behind a folder and picks one of them in secret. Ask yes-or-no questions until you are sure which block it is. Take turns, and write down how many questions each round takes.



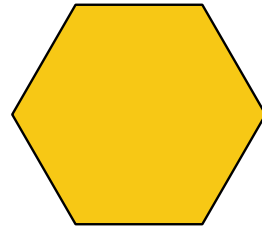
green triangle



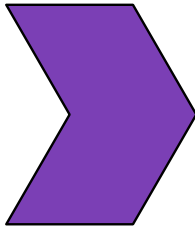
blue rhombus



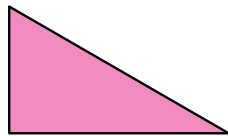
red trapezoid



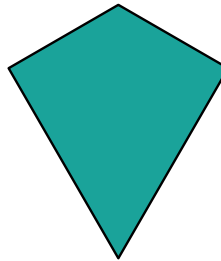
yellow hexagon



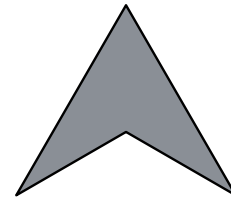
purple chevron



pink right triangle



teal kite



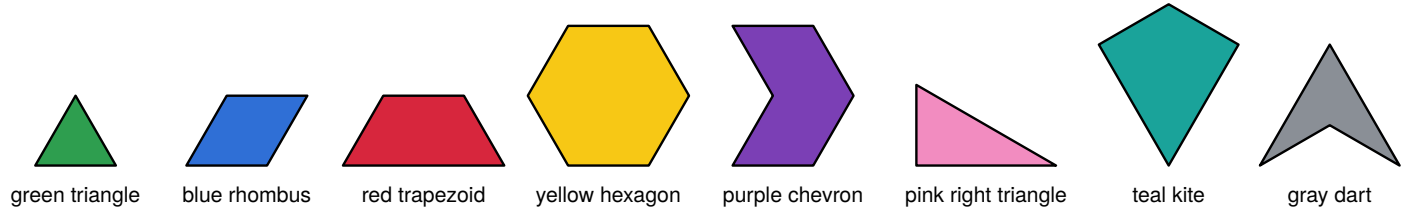
gray dart

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Then play some rounds where every question must name one block, such as “Is it the teal kite?” What is the most questions one of these rounds can take?

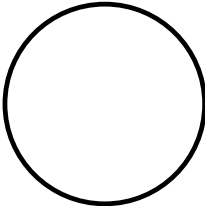
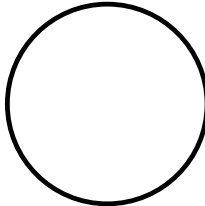
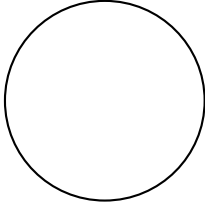
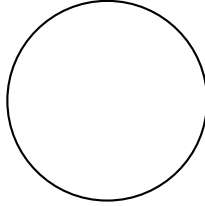
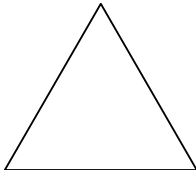
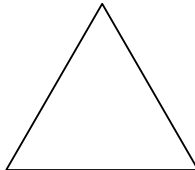
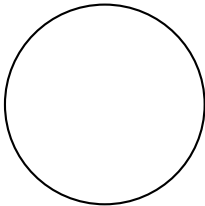
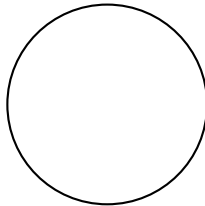
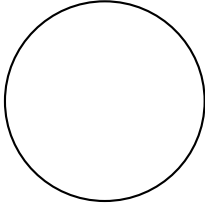
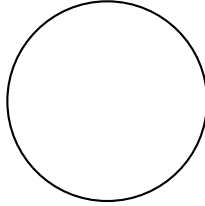
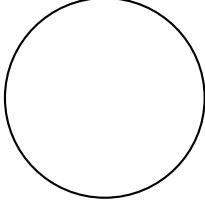
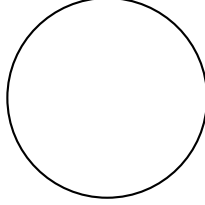
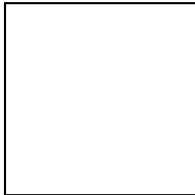
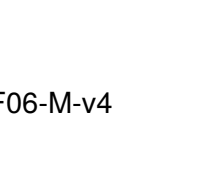
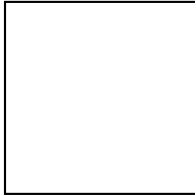
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Problem 2: Find yes-or-no questions that always tell you your partner's block in three questions or fewer. Write your questions so that someone else could use them.



Problem 3: Can two yes-or-no questions always be enough to be sure which block it is? Explain.

Problem 4: Play this code game with a partner, taking turns as keeper and breaker. The keeper hides a secret behind the folder: two counters, each red or yellow. The breaker builds a test in the next test row, with a red or yellow counter in every slot. The keeper copies the test just below the secret, puts a green triangle under each place where the two colors match, and says how many triangles there are. The breaker writes this score in the score box. When the breaker says what the secret is, the keeper lifts the folder and you check every score together. Find each secret with as few tests as you can.

secret			
copy of test			
			
			folder
test			score
			
			
			

Problem 5: Is one test always enough to be sure of a two-counter secret? Are two tests always enough? Explain.

		score

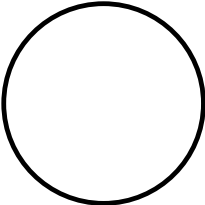
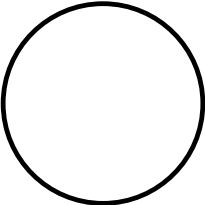
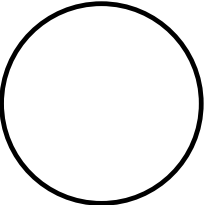
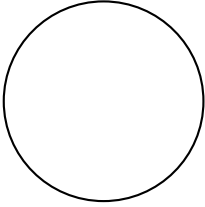
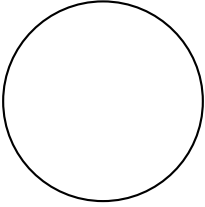
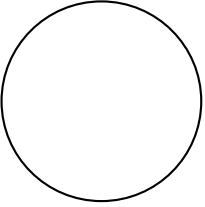
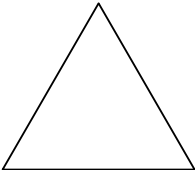
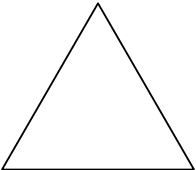
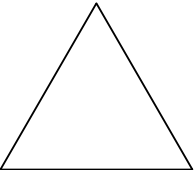
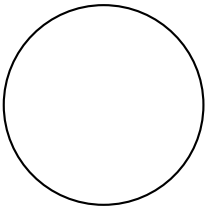
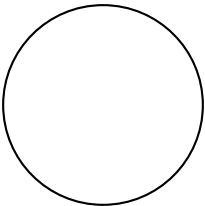
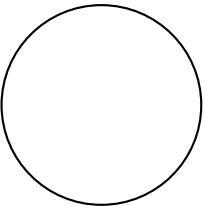
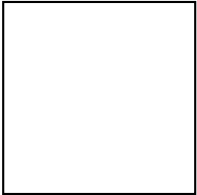
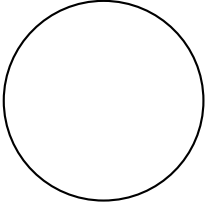
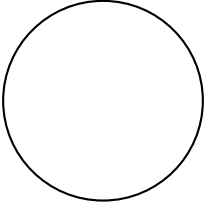
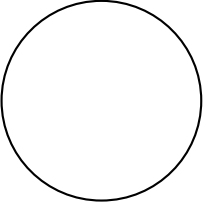
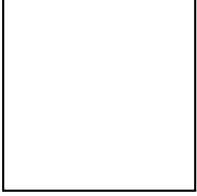
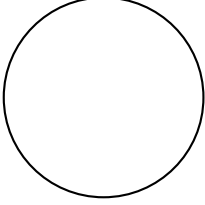
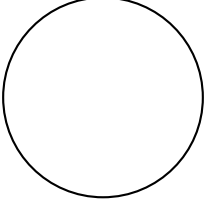
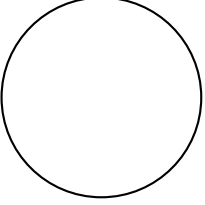
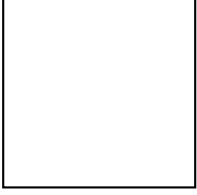
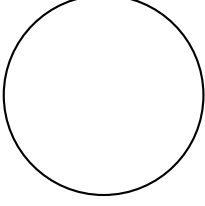
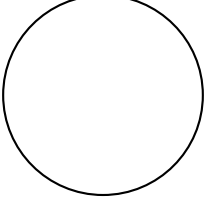
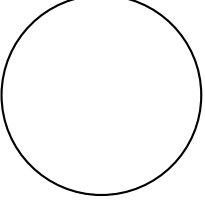
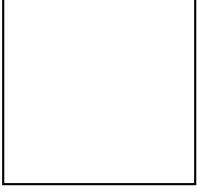
		score

		score

		score

[illegible]

Problem 6: Play the code game with three counters. Find each secret with as few tests as you can.

secret				
copy of test				
				
				folder
test				
				
				
				

Problem 7: For each of these games, write every three-counter secret that fits all the scores. If no secret fits, explain how that could happen.

			score
R	R	R	2

			score
R	Y	R	1
Y	Y	R	2

			score
R	Y	Y	2
Y	R	Y	2
R	R	R	2

			score
R	R	Y	2
R	Y	R	2
Y	R	R	2

			score
R	R	R	3
R	Y	R	1

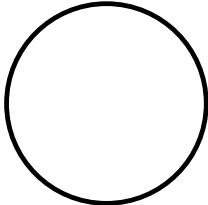
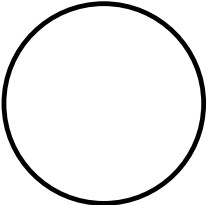
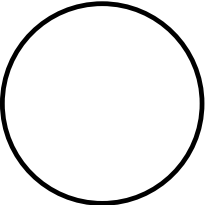
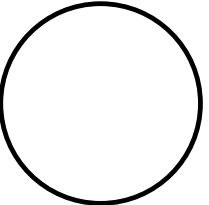
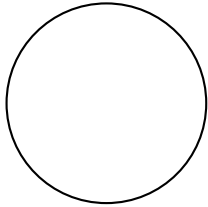
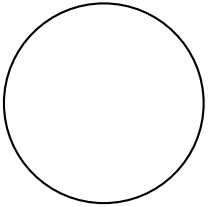
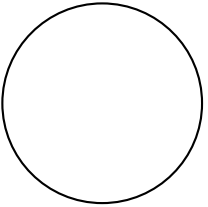
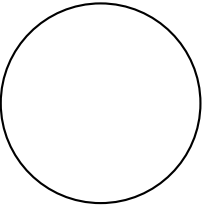
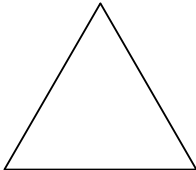
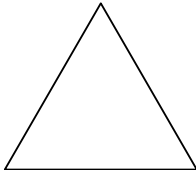
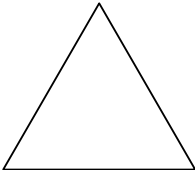
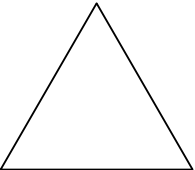
Problem 8: Find a way to choose your tests so that you always know a three-counter secret after three tests or fewer. Write it down so that someone else could use it.

[illegible][illegible][illegible]

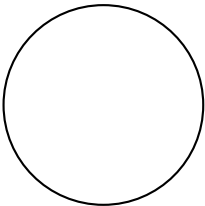
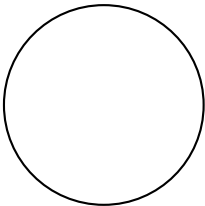
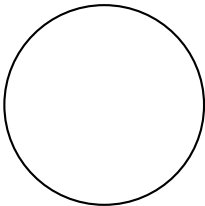
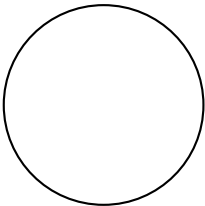
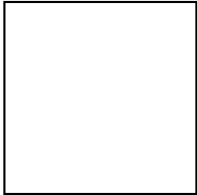
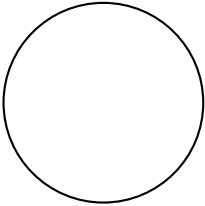
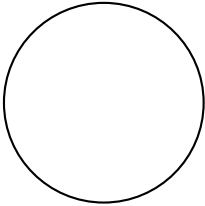
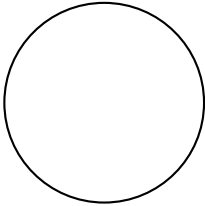
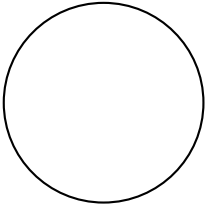
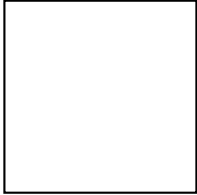
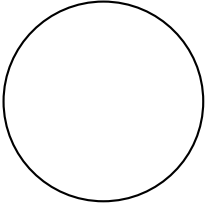
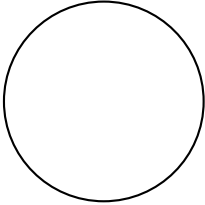
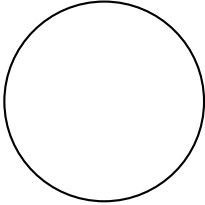
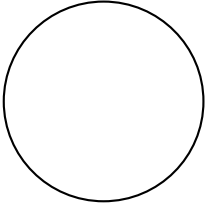
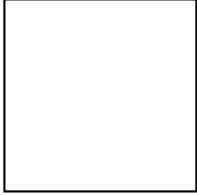
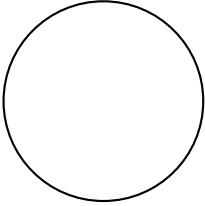
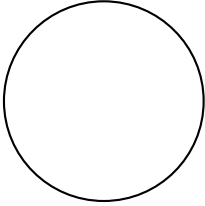
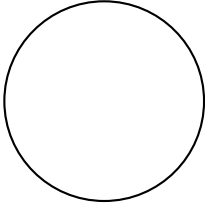
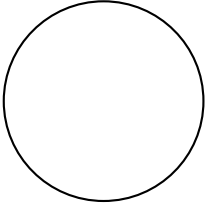
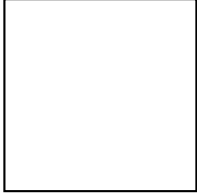
Problem 9: Play the three-counter game again, but now the game ends only when a test scores 3. Can the breaker always end the game by the fourth test? Explain.

[illegible][illegible][illegible][illegible]

Problem 10: Play the code game with four counters. Find each secret with as few tests as you can.

secret				
copy of test				
				

folder

test					
					
					
					

Problem 11: Find a way to choose your tests that always lets you know a four-counter secret. What is the most tests your way can need? Write it down so that someone else could use it.

				score						score					score	

[illegible]